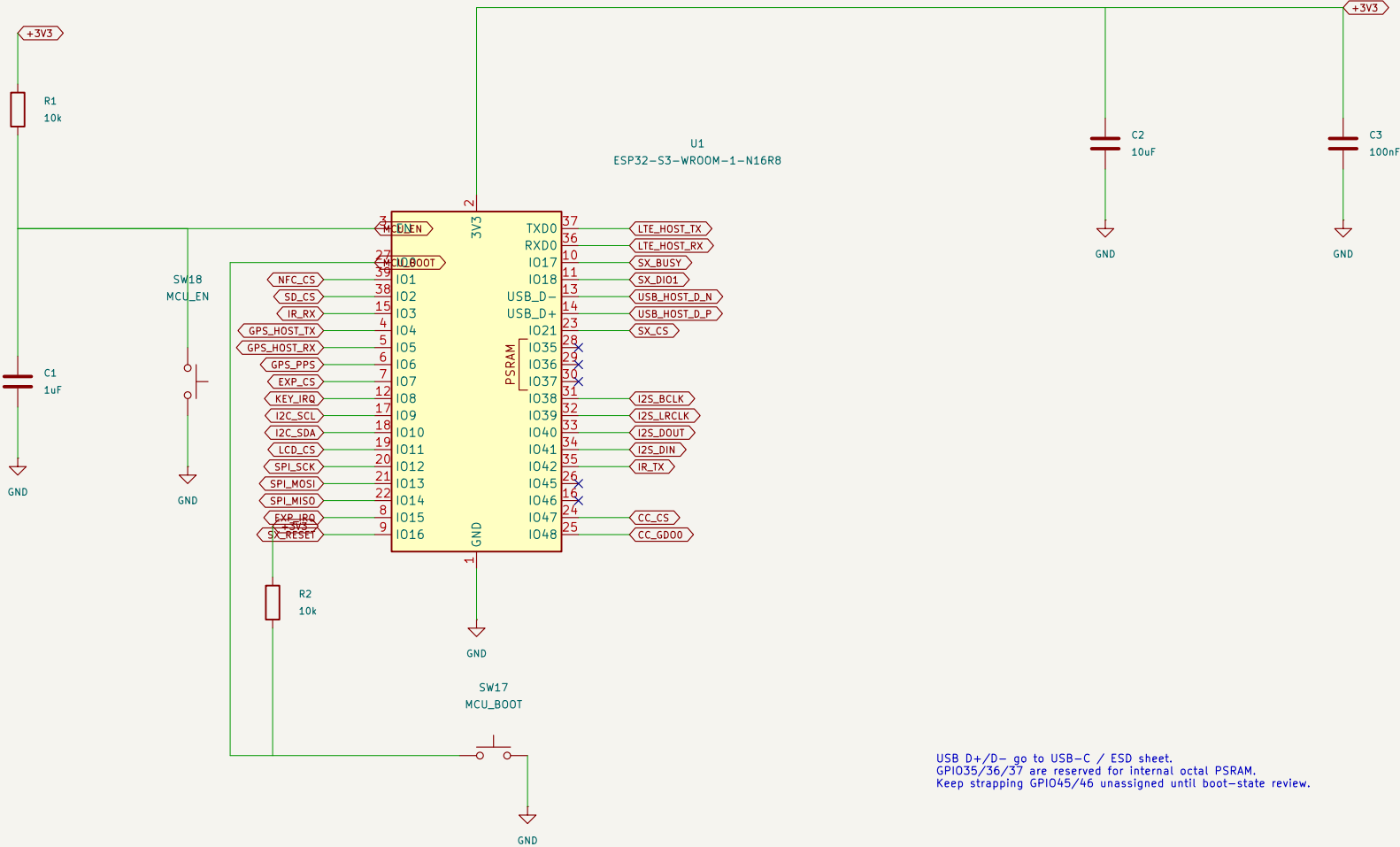


ESP32 host, boot and reset

Retain N16R8. Module pads 28/29/30 belong to octal PSRAM. USB programming uses the USB sheet.



USB D+/D- go to USB-C / ESD sheet.
GPIO35/36/37 are reserved for internal octal PSRAM.
Keep strapping GPIO45/46 unassigned until boot-state review.

RESET / EN: 10k pull-up + 1uF
BOOT held low during reset selects download mode.

Circuit review in progress – not released for fabrication

Sheet: /ESP32 host, boot and reset/
File: core.kicad_sch

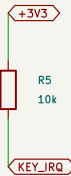
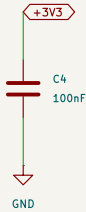
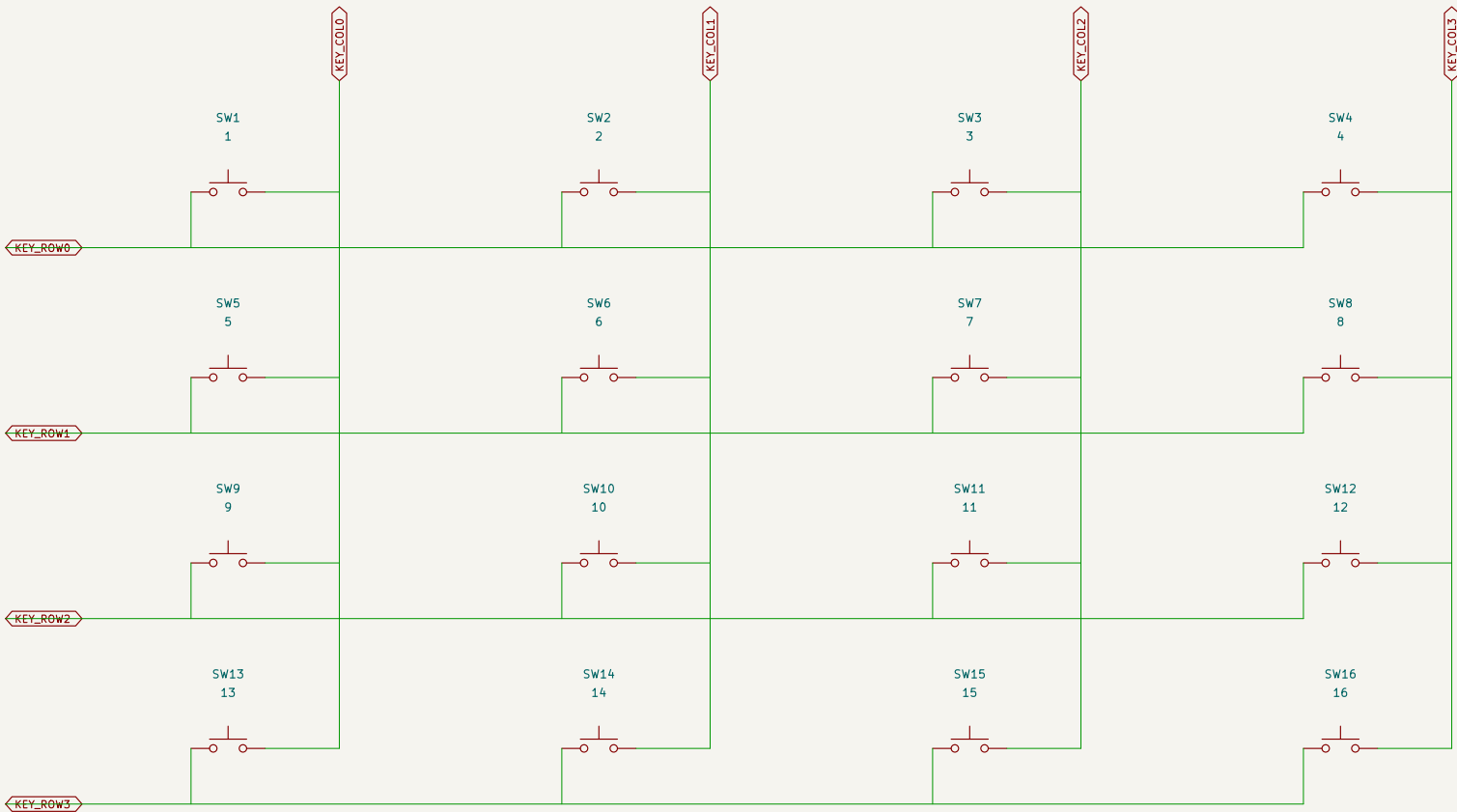
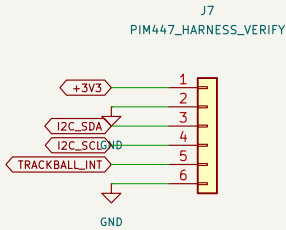
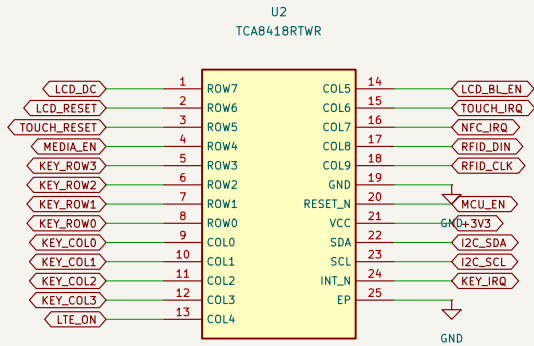
Title: ESP32 host, boot and reset

Size: A3
Date: 2026-09-24

Rev: A – engineering draft
Id: 2/25

Key matrix and trackball

TCAB418: 4 rows x 4 columns; remaining pins are firmware-configured GPIO. Confirm PIM447 harness orientation.



Circuit review in progress – not released for fabrication

Sheet: /Key matrix and trackball/
File: keys.kicad_sch

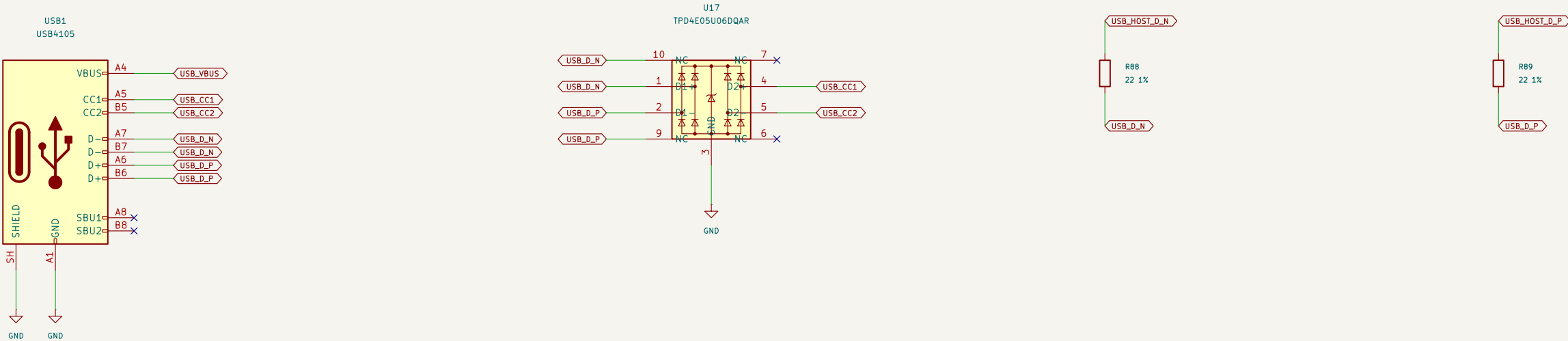
Title: Key matrix and trackball

Size: A3
Date: 2026-09-24

Rev: A – engineering draft
Id: 3/25

USB-C data and ESD

U17 NC lands 10/9 externally bridge to 1/2 for pass-through routing; no internal connection. R88/R89 host series termination, initial 22 ohms; qualify on hardware.



Circuit review in progress – not released for fabrication

Sheet: /USB-C data and ESD/
File: usb.kicad_sch

Title: USB-C data and ESD

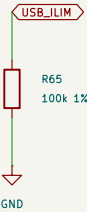
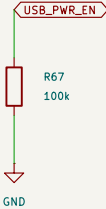
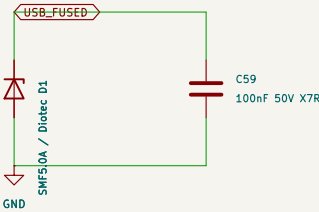
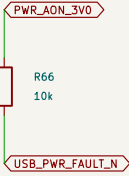
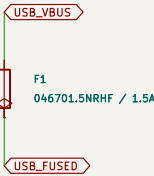
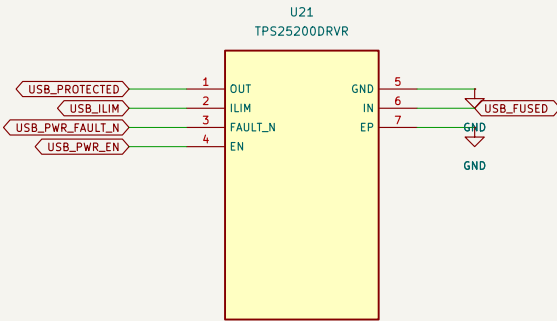
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 4/25

Fused USB input and eFuse

TPS25200 R_ILIM = 100k, 1%. F1/TVS pulse coordination requires bench validation. No USB PD negotiation.



Supply flag: external USB VBUS through F1. No internal source is implied.

Circuit review in progress – not released for fabrication

Sheet: /Fused USB input and eFuse/
File: usb-input.kicad_sch

Title: Fused USB input and eFuse

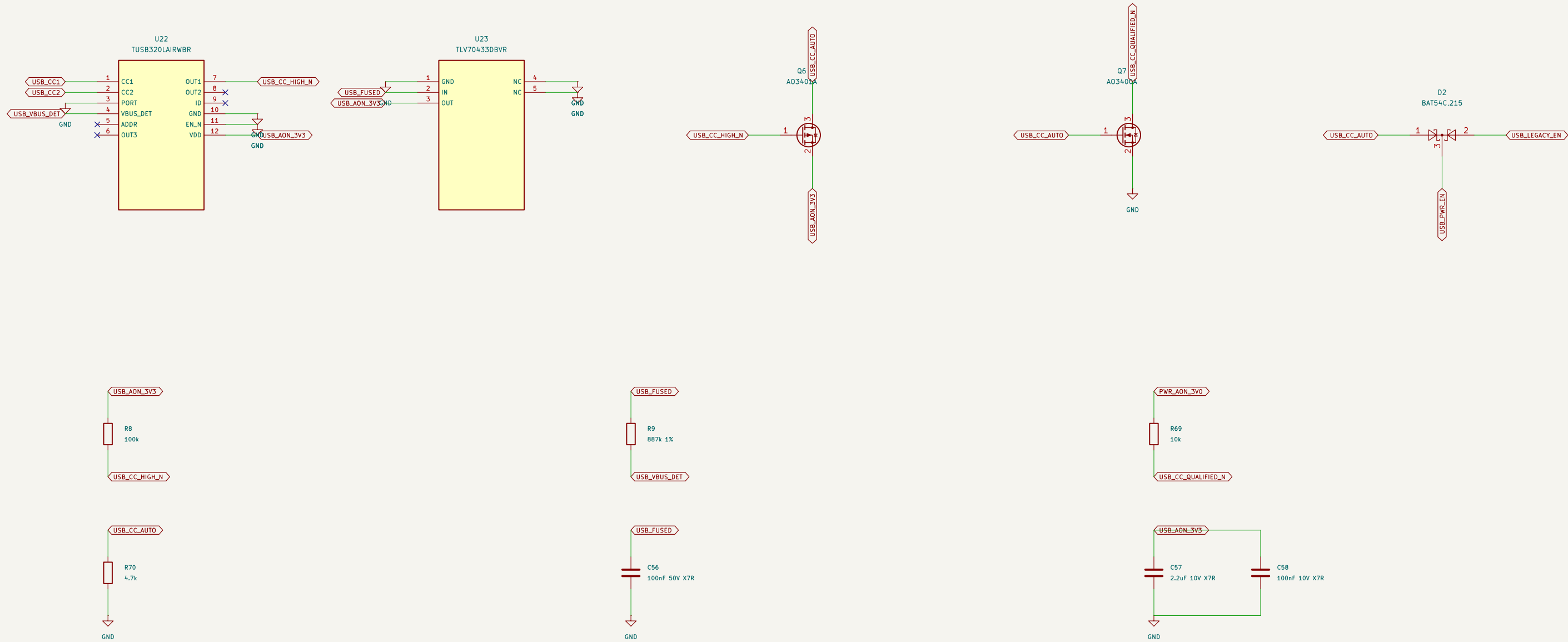
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KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 5/25

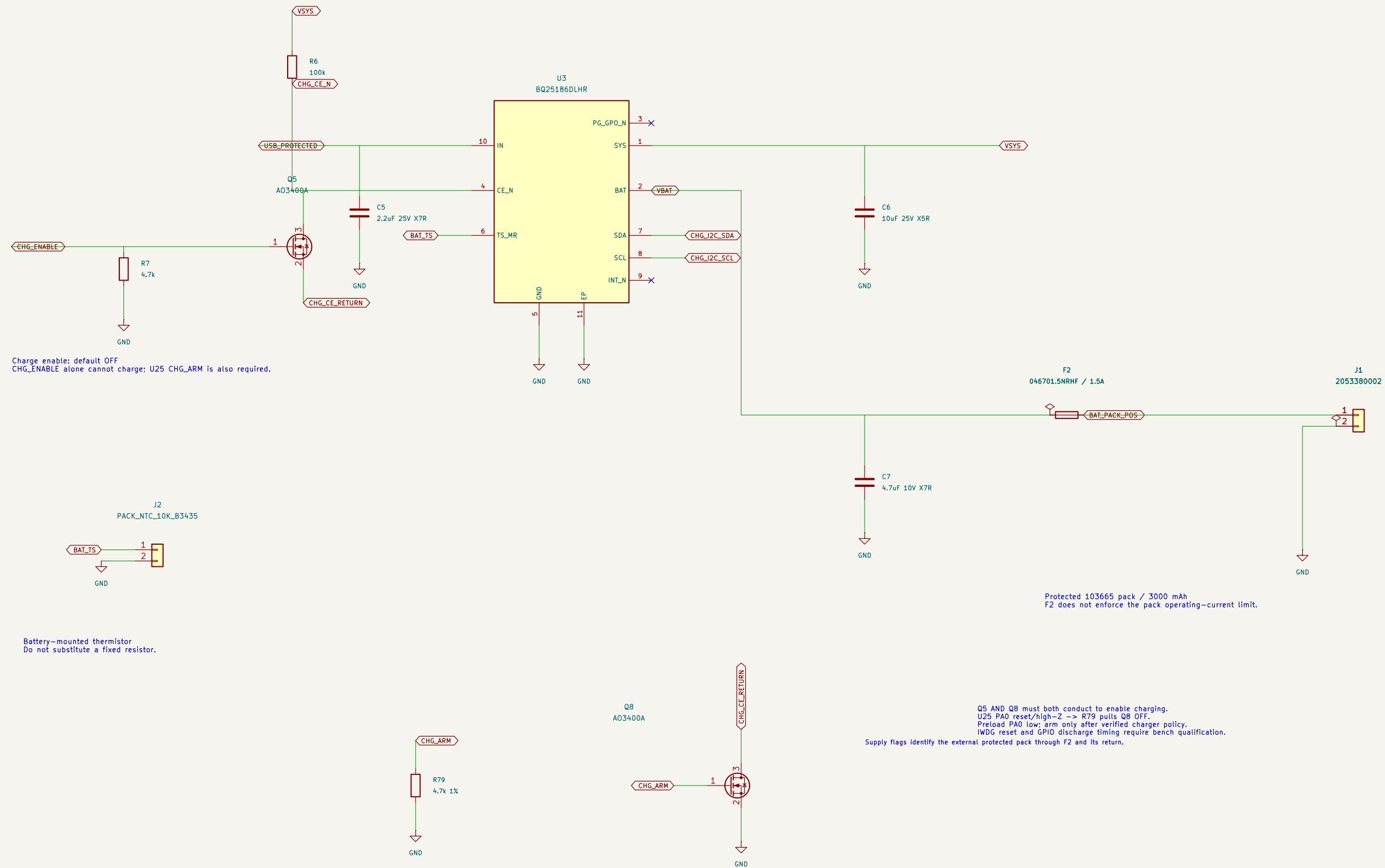
USB-C source detection and permission

Sink-only internal Rd. OUT1 low authorizes Type-C 1.5A/3A. Legacy permission requires USB enumeration; see charger-policy.json.



Battery, thermistor and programmable charger

Verify actual battery polarity. J2 is a battery-mounted 10k B3435 NTC. CE defaults OFF. Firmware must apply charger-policy.json.



Circuit review in progress – not released for fabrication

Sheet: /Battery, thermistor and programmable charger/
File: power.kicad_sch

Title: Battery, thermistor and programmable charger

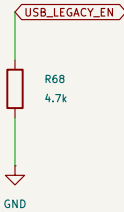
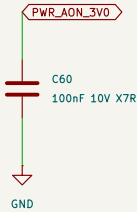
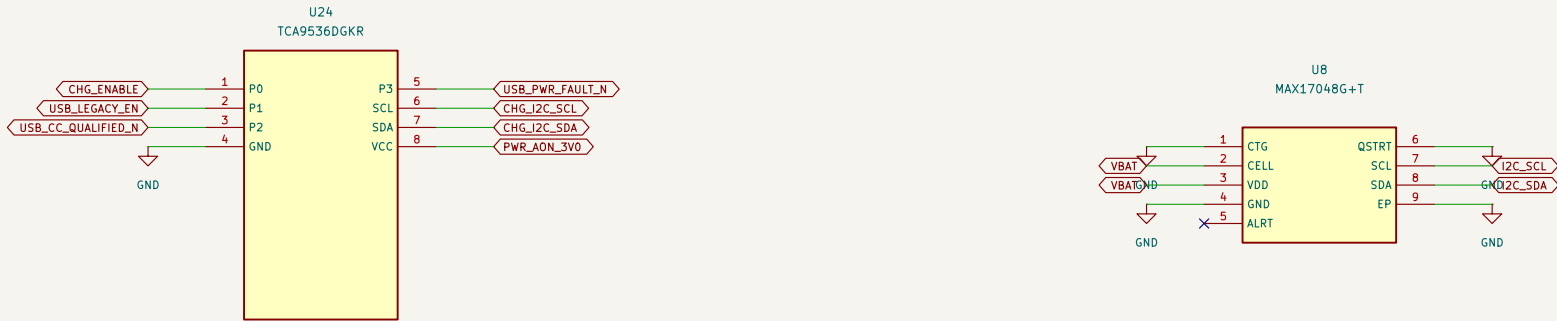
Size: A3	Date: 2026-09-24
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Rev: A – engineering draft

Size: A5	D
KiCad E.D.A. 10.0.6	

Power control and fuel gauge

I2C: TCA9536 0x41, MAX17048 0x36, charger 0x6A. Preload low output latches before enabling charger—control outputs.



Circuit review in progress – not released for fabrication

Sheet: /Power control and fuel gauge/
File: power-control.kicad_sch

Title: Power control and fuel gauge

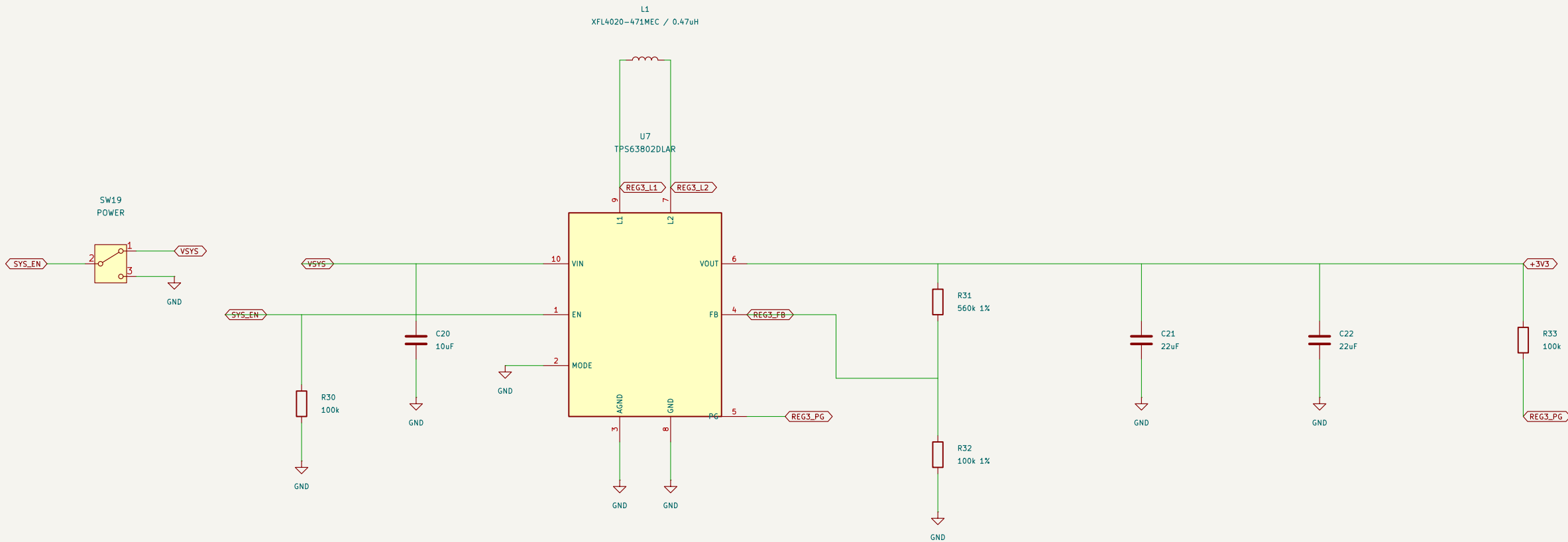
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KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 8/25

Main 3.3 V buck–boost supply

TPS63802 local PCB copper is routed. External distribution, effective capacitance, thermal limits and load transients remain to qualify.



0.6 V reference x (1 + 560k / 100k) = 3.3 V nominal.
Feedback senses the output capacitor node; keep it clear of the switching loop.

SW19 controls SYS_EN.
Battery / charger circuitry can remain energized while logic rails are off.

Circuit review in progress – not released for fabrication

Sheet: /Main 3.3 V buck–boost supply/
File: reg3.kicad_sch

Title: Main 3.3 V buck–boost supply

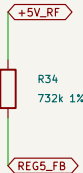
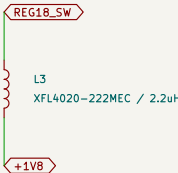
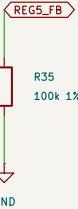
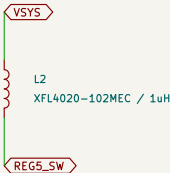
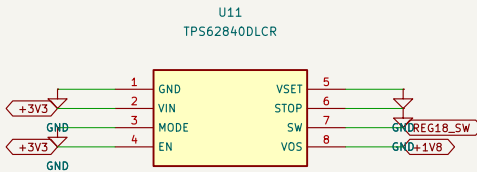
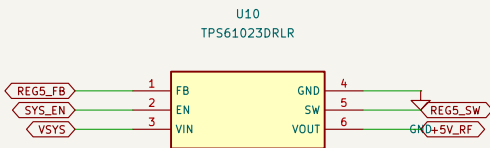
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026–09–24

Rev: A – engineering draft
Id: 9/25

Auxiliary 5 V and 1.8 V supplies

Switching loops must be short in PCB layout. The 5 V rail also supplies display backlight and the LF reader.



Circuit review in progress – not released for fabrication

Sheet: /Auxiliary 5 V and 1.8 V supplies/
File: aux-power.kicad_sch

Title: Auxiliary 5 V and 1.8 V supplies

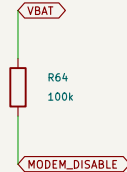
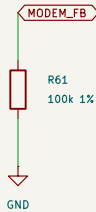
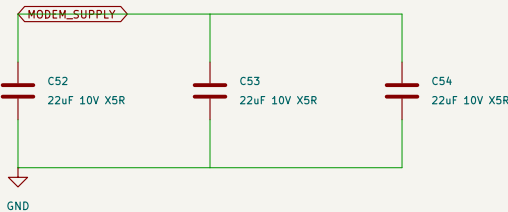
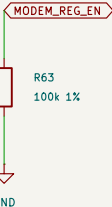
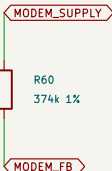
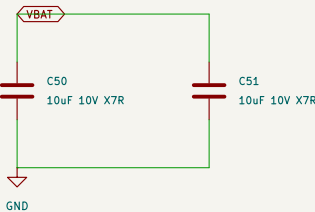
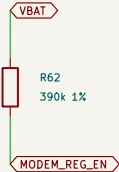
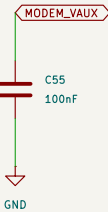
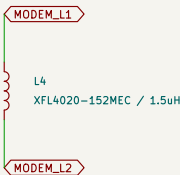
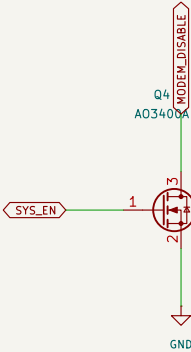
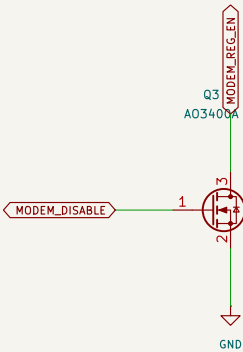
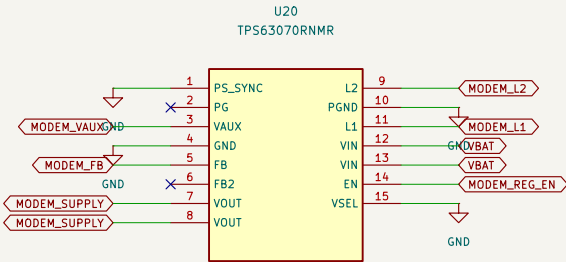
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 10/25

Regulated modem supply and enable

Nominal output 3.792 V. Exact modem burst current and MakerFocus pack current budget are unresolved release blockers.



Circuit review in progress – not released for fabrication

Sheet: /Regulated modem supply and enable/
File: modem-power.kicad_sch

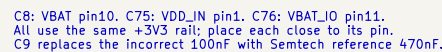
Title: Regulated modem supply and enable

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 11/25

SX1262 clock, supply, PA feed and RF matching/switch captured. RF layout incomplete; do not power.



Y3 pads 1/2 = GND, 3 = OUT, 4 = VDD. XTB is intentionally NC.
Configure DIO3 for 1.8V; provisional startup allowance 5ms.
Confirm TCXO output $\leq 1.2\text{Vpp}$; no qualified maximum in series data.

U37: CTRL=0 selects RF1/RX; CTRL=1 selects RF2/TX.
R93 defaults to RX. C98/C99/C100 explicitly block DC.
100pF RF blocks are starting values, subject to RF measurement.
No RF route or thermal/ground-via qualification in this checkpoint.

SetRegulatorMode(0x01) only in STDBY_RC, after hardware qualification.
L12 supplies the core, not the SX1262 high-power PA.
L13: 47nH PA choke; C10/C101: 47nF/47pF bypass.
RAK4270 values; final RF layout and qualification remain open.

Circuit review in progress – not released for fabrication

Sheet: /SX1262 LoRa radio/
File: radios.kicad_sch

Title: SX1262 LoRa radio

Size: A3	Date: 2026-10-08
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KiCad E.D.A. 10.0.6

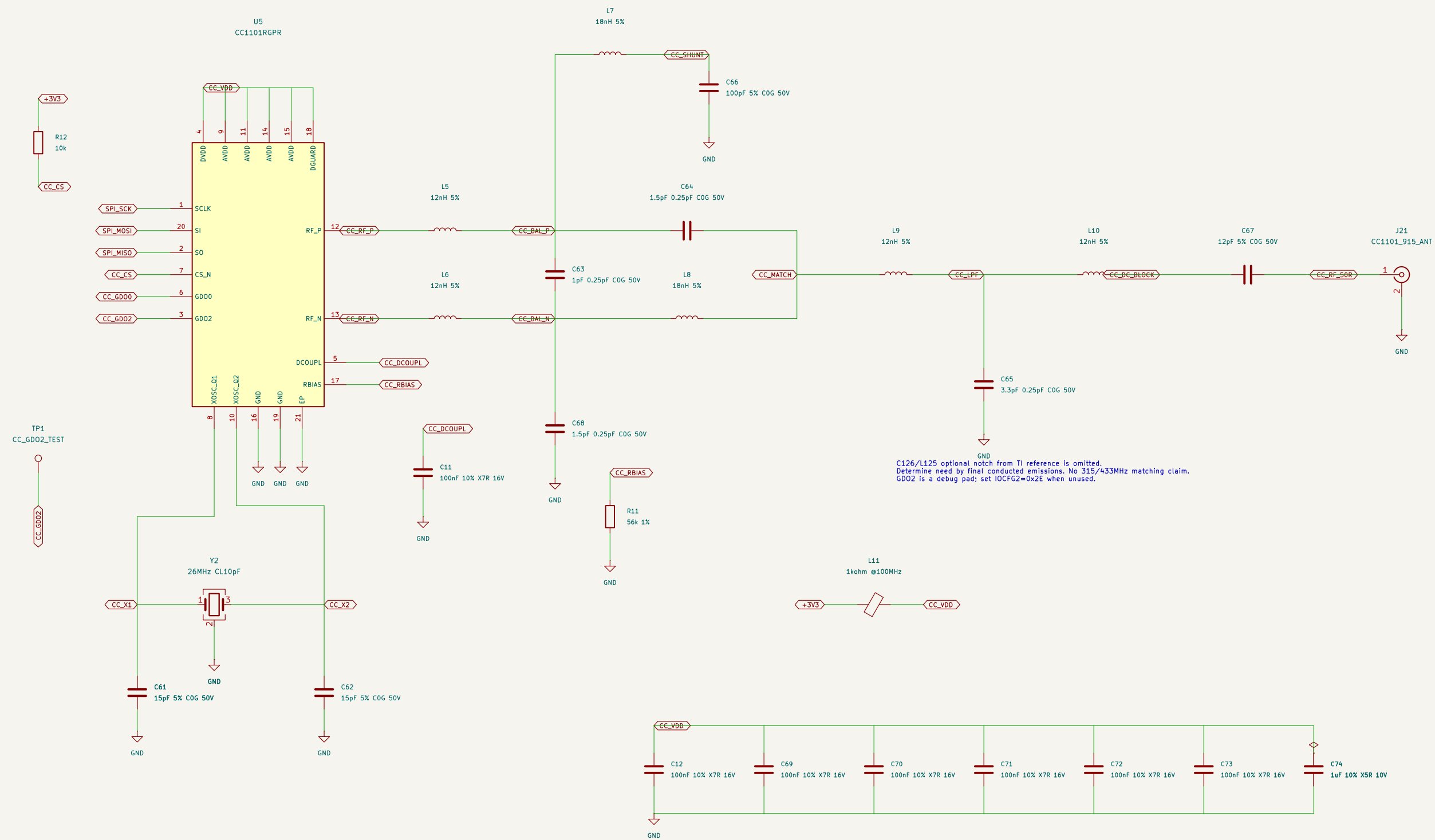
Date: 2026-10-08

Rev: A – engineering draft

Id: 12/25

CC1101 sub-GHz radio

868/915MHz frontend: TI SWRR045 rev3.0 / SWRS061I Fig11. RF layout and tuning are NOT qualified. Separate 50-ohm antenna.



Y2: ABM8-26.000MHZ-10-D-1-G-T
CL = 15pF / 2 + estimated 2.5pF stray = 10pF.
Measure oscillator frequency, startup and drive level.

CC_VDD flag: main +3V3 through passive L11; not an independent source.
Place one 100nF at each supply pin group, plus local 1uF bulk.

Circuit review in progress – not released for fabrication

Sheet: /CC1101 sub-GHz radio/
File: cc-radio.kicad_sch

Title: CC1101 sub-GHz radio

Size: A3	Date: 2026-09-24
KiCad E.D.A. 10.0.6	

Rev: A – engineering draft
Id: 13/25

125 kHz reader analog circuit

HTRC110 requires its oscillator, QGND/CEXT bypass, resonant coil network and a timing-capable host interface.

Engineering draft / named nets are common / NC = intentional unused pin

Circuit review in progress – not released for fabrication		
Sheet: /125 kHz reader analog circuit/ File: lf-radio.kicad_sch		
Title: 125 kHz reader analog circuit		
Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6		Id: 14/25

125 kHz reader analog circuit

HTRC110 requires its oscillator, QGND/CEXT bypass, resonant coil network and a timing-capable host interface.

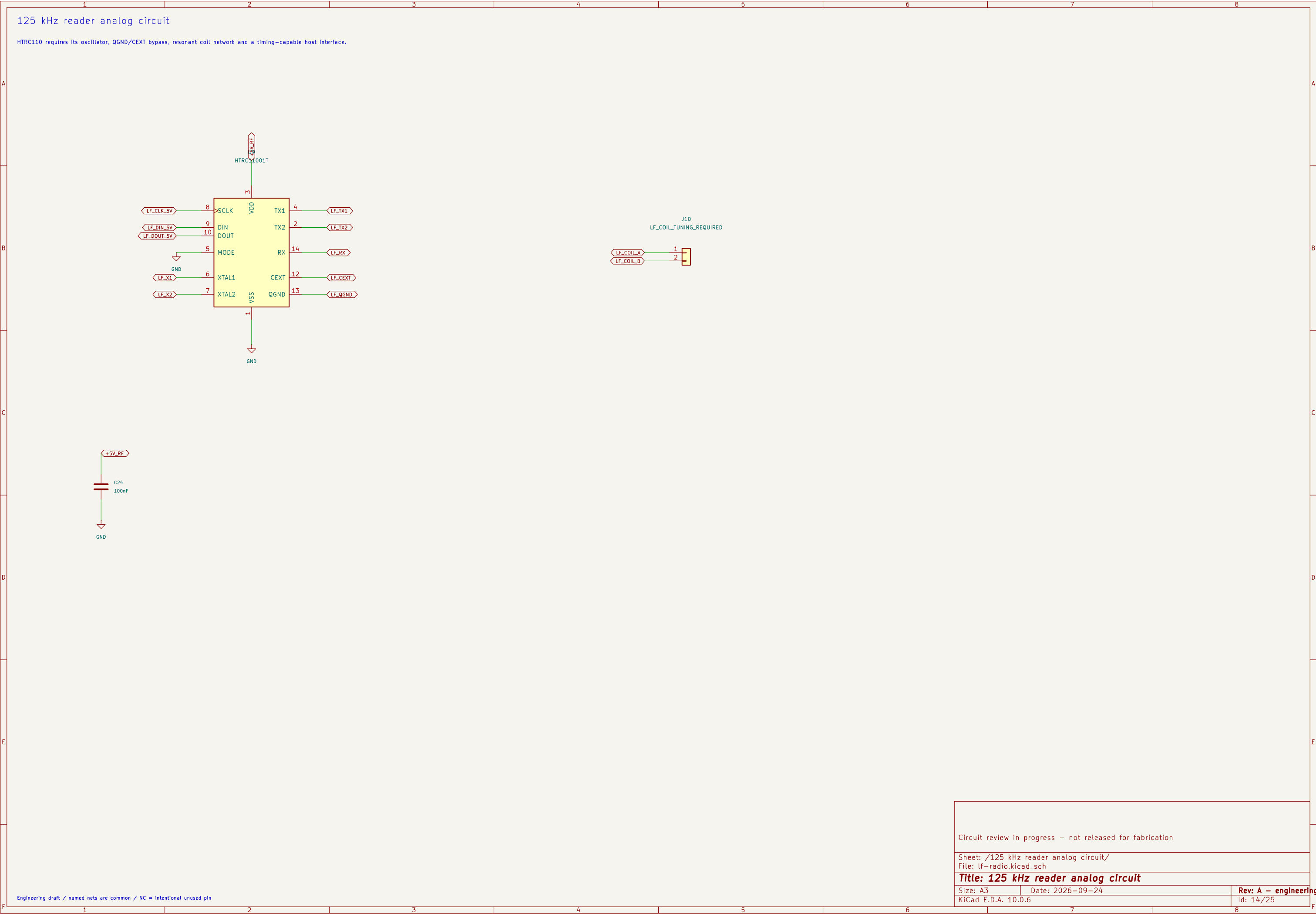
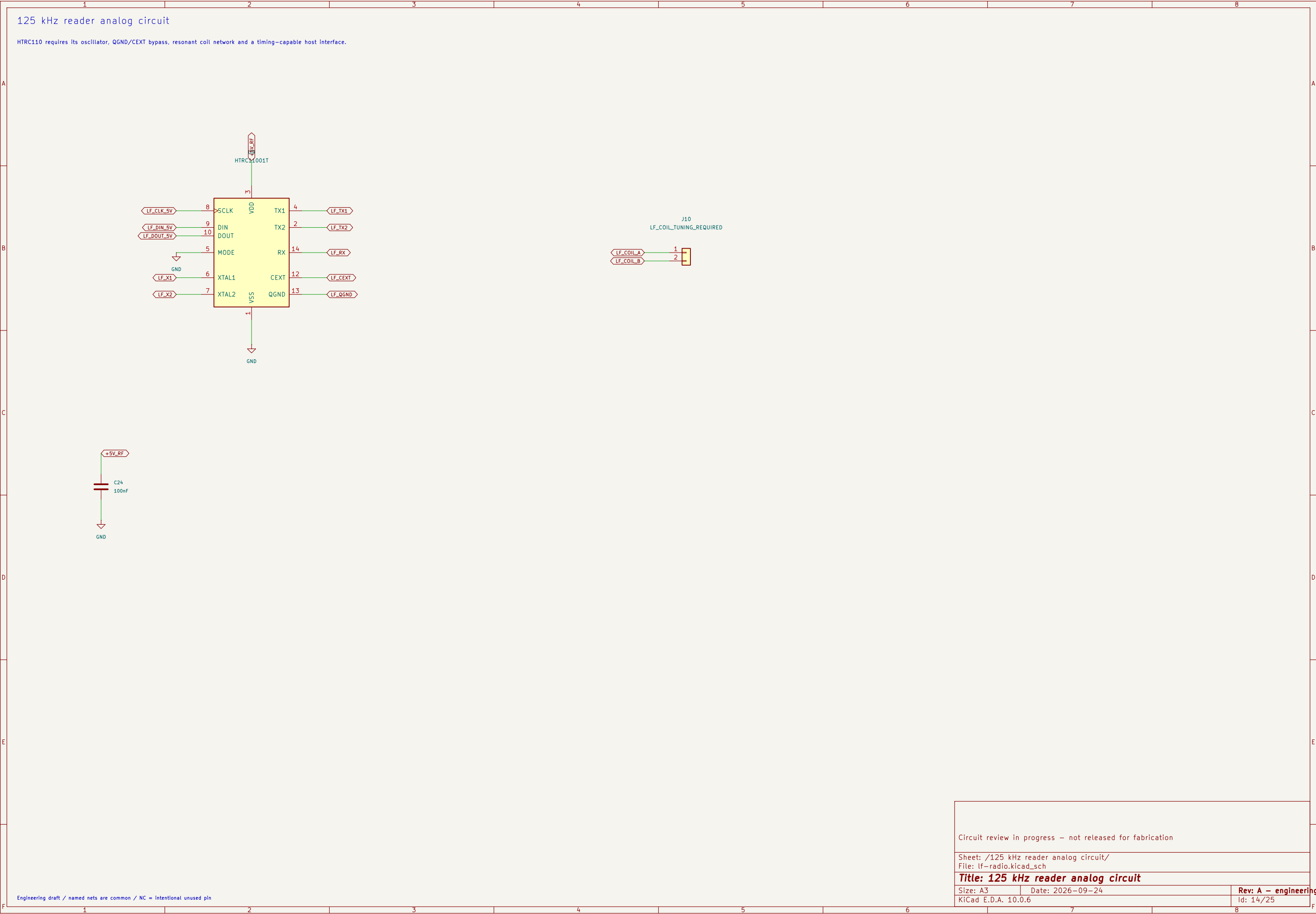
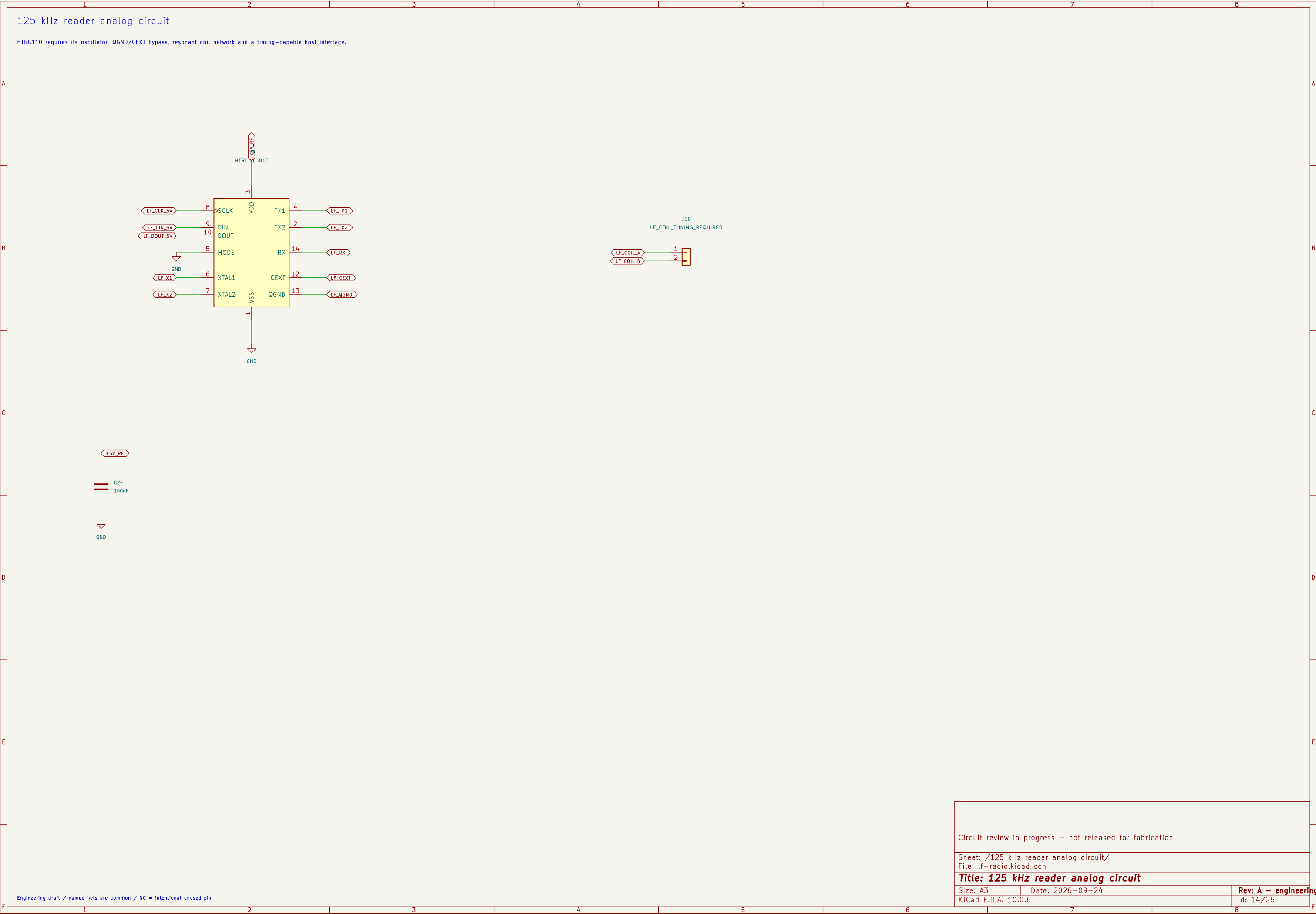
The schematic diagram illustrates the 125 kHz reader analog circuit. The central component is the HTRC1101T IC, which is connected to various pins. Pin 1 (VSS) is connected to GND. Pin 2 (VDD) is connected to +5V_RF. Pin 3 (HTRC1101T) is connected to GND. Pin 4 (TX1) is connected to LF_TX1. Pin 5 (MODE) is connected to GND. Pin 6 (XTAL1) is connected to LF_X1. Pin 7 (XTAL2) is connected to LF_X2. Pin 8 (SCLK) is connected to LF_CLK_5V. Pin 9 (DIN) is connected to LF_DIN_5V. Pin 10 (DOUT) is connected to LF_DOUT_5V. Pin 11 (TX2) is connected to LF_TX2. Pin 12 (CEXT) is connected to LF_CEXT. Pin 13 (QGND) is connected to LF_QGND. Pin 14 (RX) is connected to LF_RX. A capacitor C24 (100nF) is connected between +5V_RF and GND. A coil network J10 (LF_COIL_TUNING_REQUIRED) is connected between LF_COIL_A and LF_COIL_B.

Circuit review in progress – not released for fabrication

Sheet: /125 kHz reader analog circuit/
File: lf-radio.kicad_sch

Title: 125 kHz reader analog circuit

Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6		Id: 14/25



Sheet: /125 kHz reader analog circuit/ File: lf-radio.kicad_sch		
Title: 125 kHz reader analog circuit		
Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6	Id: 14/25	

Sheet: /125 kHz reader analog circuit/ File: lf-radio.kicad_sch		
Title: 125 kHz reader analog circuit		
Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6	Id: 14/25	

Title: 125 kHz reader analog circuit		
Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6		Id: 14/25

Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6		Id: 14/25

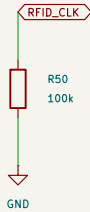
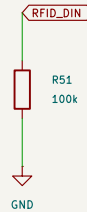
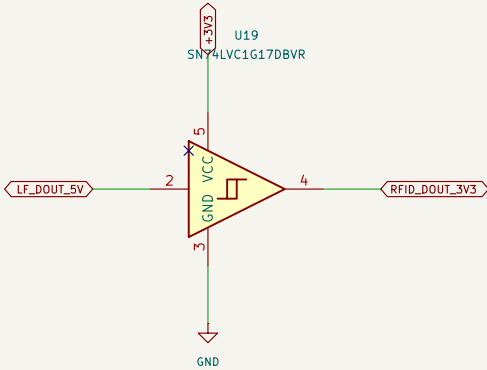
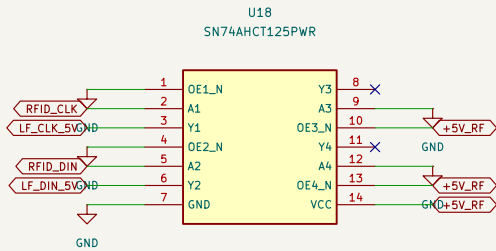
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KiCad E.D.A. 10.0.6		Id: 14/25

Size: A3	Date: 2026-09-24	Rev: A – engineering
KiCad E.D.A. 10.0.6		Id: 14/25

DATE: 2025-05-27	REV: 1
KiCad E.D.A. 10.0.6	Id: 14/25

LF reader logic translation

AHCT inputs accept 3.3 V logic at 5 V supply; LVC return buffer is powered at 3.3 V. Host timing remains a design requirement.



Circuit review in progress – not released for fabrication

Sheet: /LF reader logic translation/
File: lf-support.kicad_sch

Title: LF reader logic translation

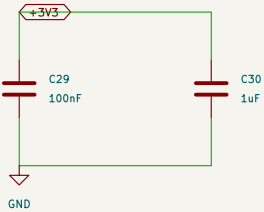
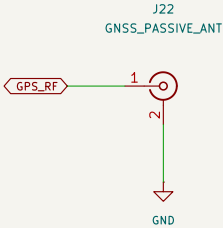
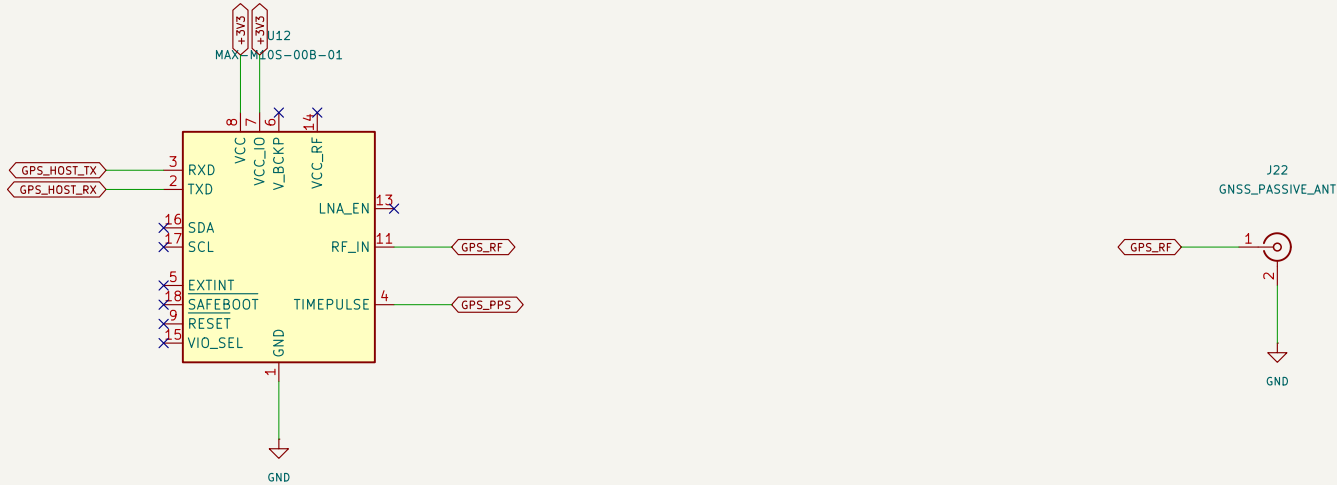
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 15/25

GNSS receiver and passive antenna

MAX-M10S passive antenna. V_BCKP remains open: cold starts after power removal. No direct LiPo connection.



Circuit review in progress – not released for fabrication

Sheet: /GNSS receiver and passive antenna/
File: gnss-antennas.kicad_sch

Title: GNSS receiver and passive antenna

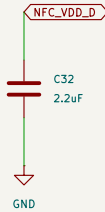
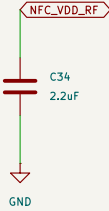
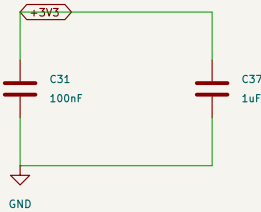
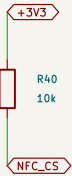
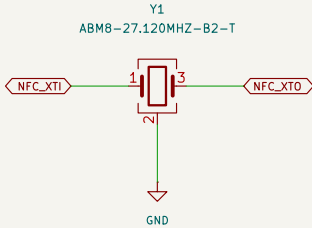
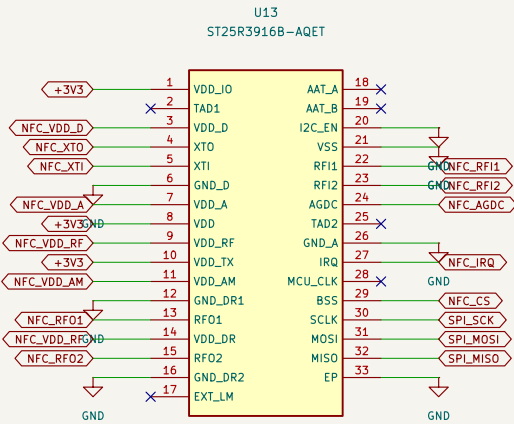
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 16/25

NFC reader, oscillator and local supplies

ST25R3916B needs a tuned case-mounted loop and TX/RX matching network. Verify crystal load against the selected crystal.



Circuit review in progress – not released for fabrication

Sheet: /NFC reader, oscillator and local supplies/
File: nfc.kicad_sch

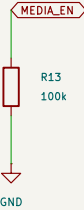
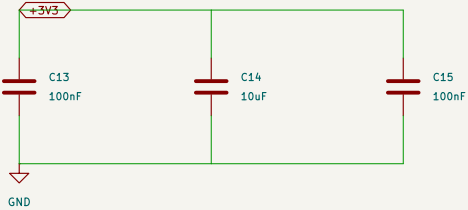
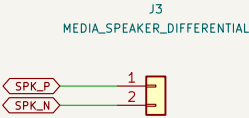
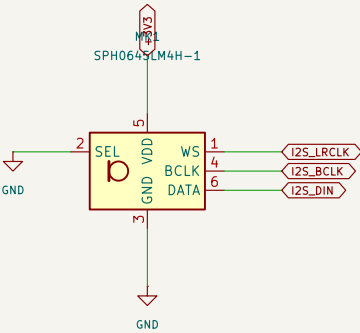
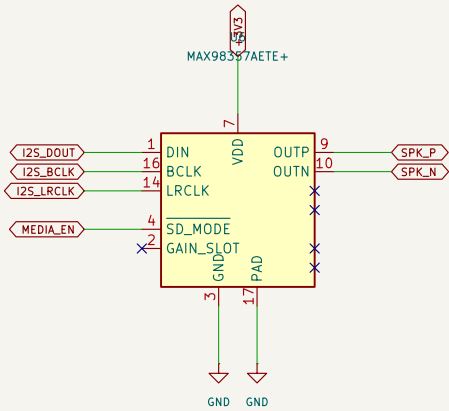
Title: NFC reader, oscillator and local supplies

Size: A3
Date: 2026-09-24

Rev: A – engineering draft
Id: 17/25

Media speaker amplifier and digital microphone

Top-mounted media speaker uses differential outputs: neither speaker wire is ground. Digital microphone is case-bottom mounted.



Circuit review in progress – not released for fabrication

Sheet: /Media speaker amplifier and digital microphone/
File: audio.kicad_sch

Title: Media speaker amplifier and digital microphone

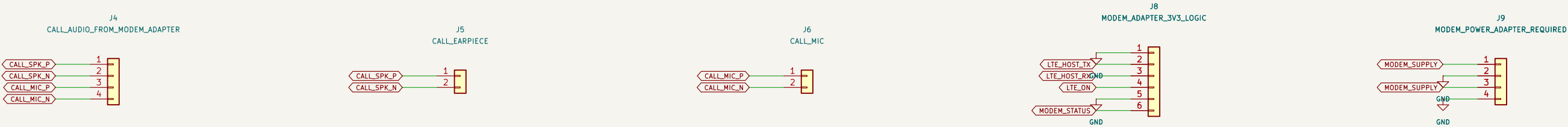
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 18/25

Cellular adapter and call–audio harnesses

Replaceable modem adapter owns SIM, cellular RF and modem–level translation. No universal–carrier guarantee. Speaker top / mic bottom.



Circuit review in progress – not released for fabrication

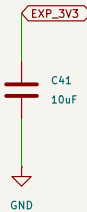
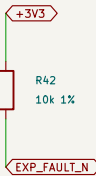
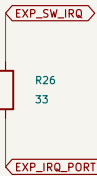
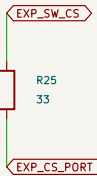
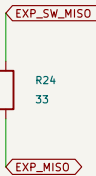
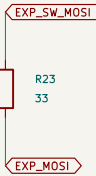
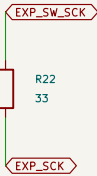
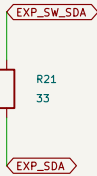
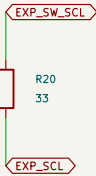
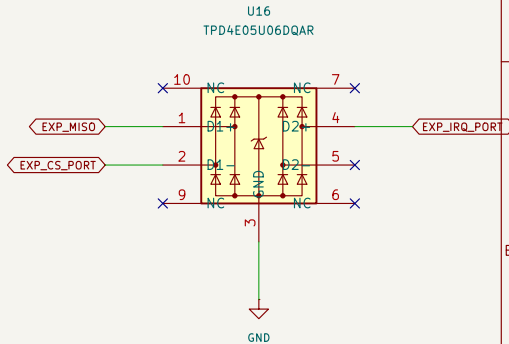
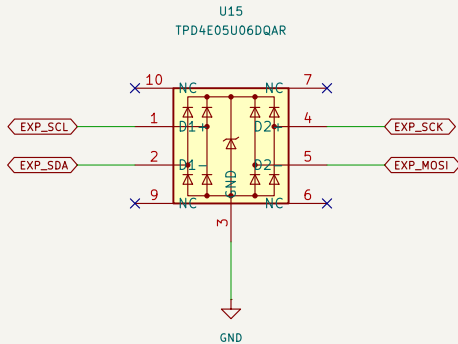
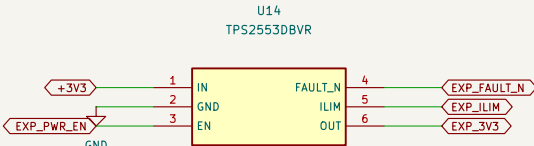
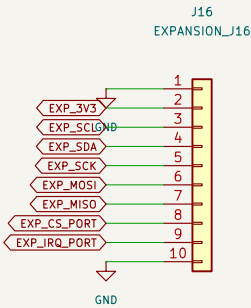
Sheet: /Cellular adapter and call–audio harnesses/
File: interfaces.kicad_sch

Title: Cellular adapter and call–audio harnesses

Size: A3	Date: 2026–09–24	Rev: A – engineering draft
KiCad E.D.A. 10.0.6		Id: 19/25

Protected external module port

Port signals pass through U34 on expansion-control. U14 EN defaults low; fault drives U33 and the hardware signal interlock.
No hot-plug approval.



Circuit review in progress – not released for fabrication

Sheet: /Protected external module port/
File: protection.kicad_sch

Title: Protected external module port

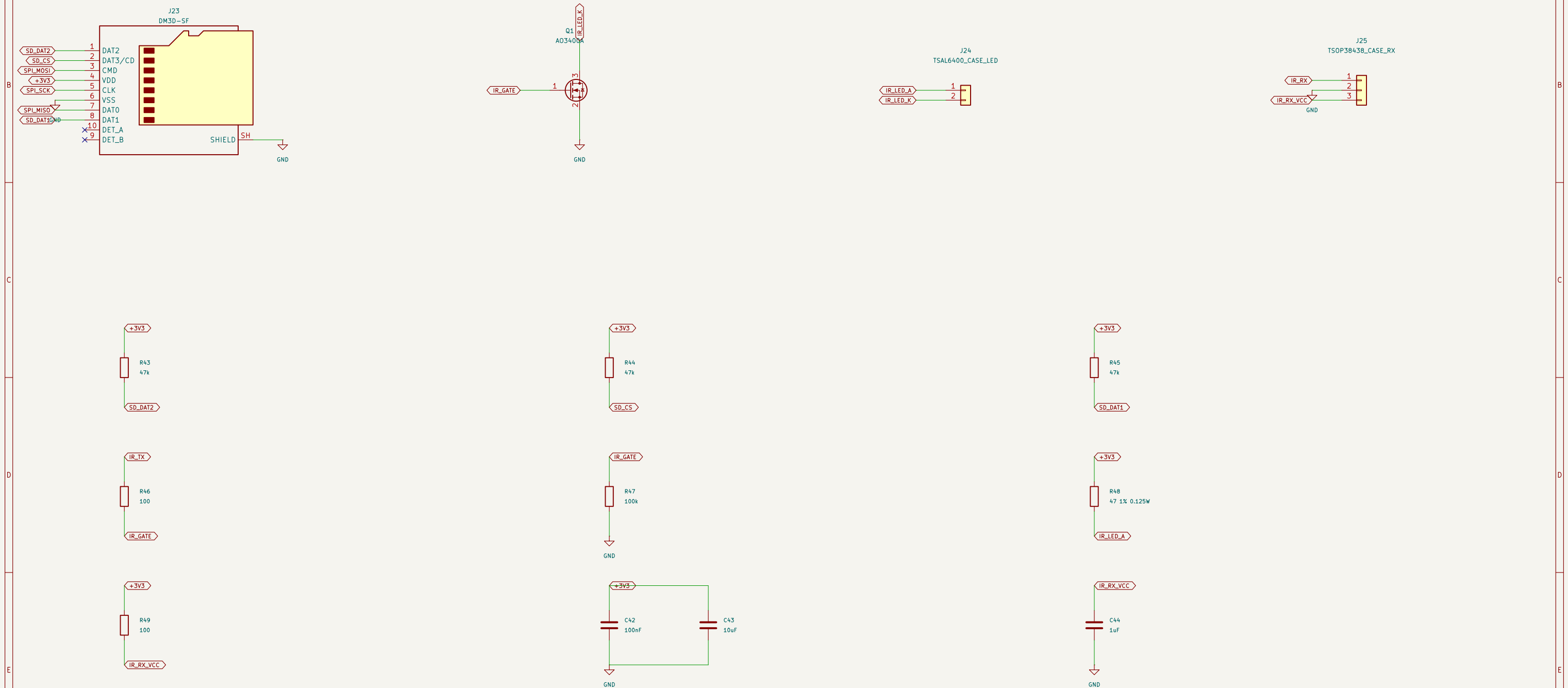
Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-24

Rev: A – engineering draft
Id: 20/25

MicroSD and case-mounted infrared

SPI microSD, transistor-driven IR LED, filtered receiver supply. Confirm case LED/receiver polarity and GPIO3 boot behavior.



Circuit review in progress – not released for fabrication

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File: storage-ir.kicad_sch

Title: MicroSD and case-mounted infrared

Size: A3

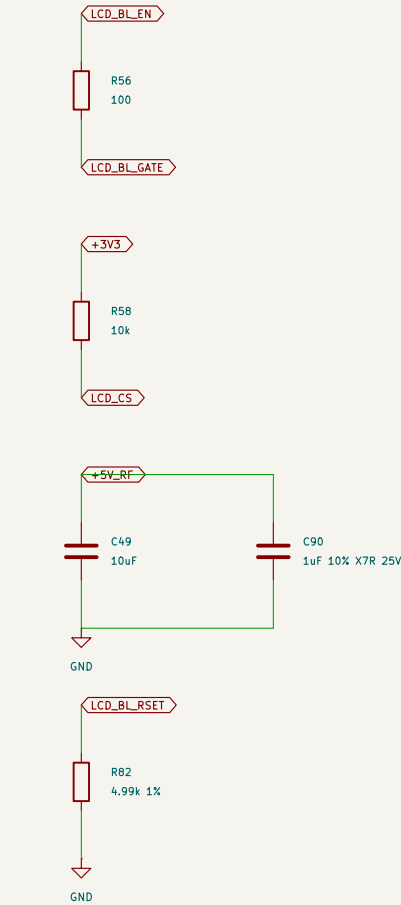
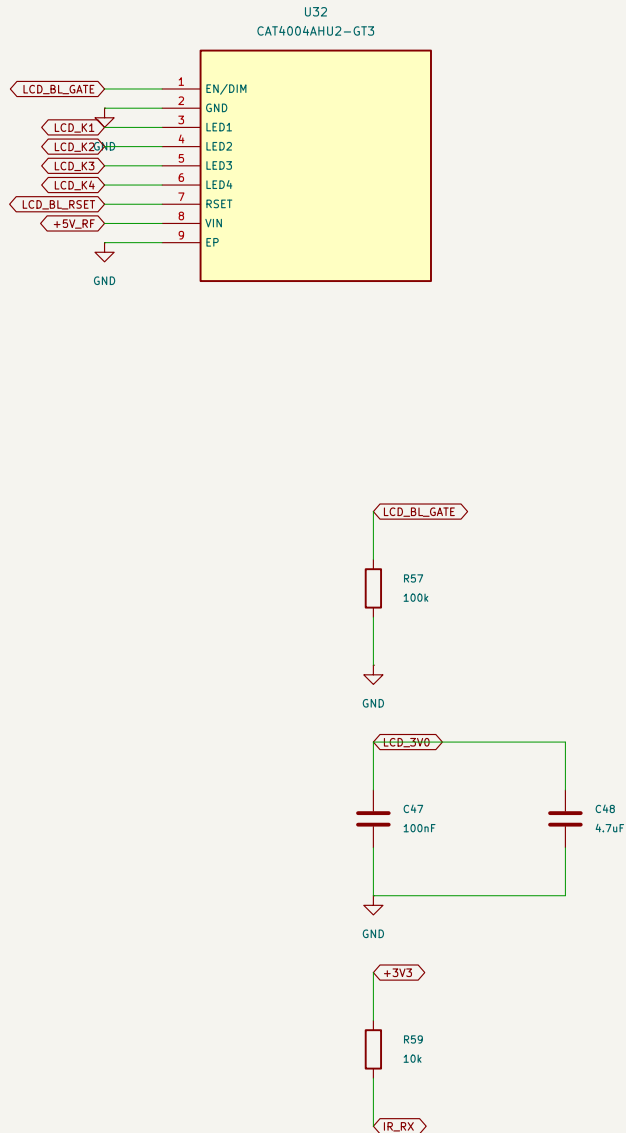
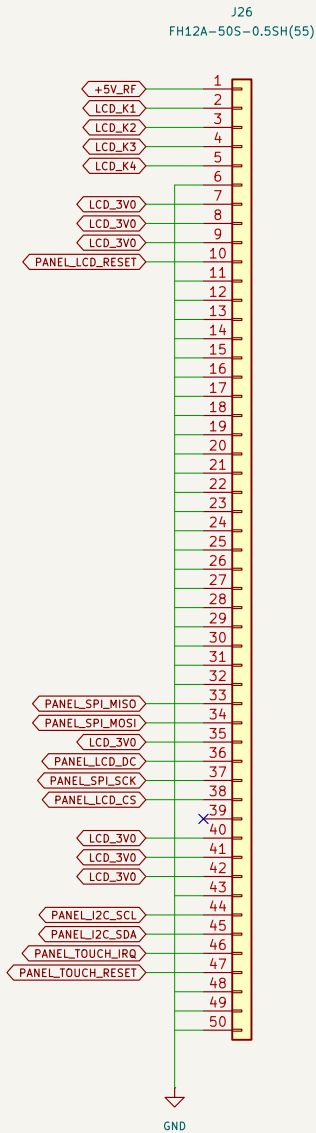
Date: 2026-09-24

Rev: A – engineering draft

Id: 21/25

External display, touch and backlight

CAT4004A: four current sinks, 4.99k sets approx. 15mA/channel. On/off only until dimming timing is qualified. Do not fabricate.



EN/DIM: external R57 pull-down retains default off.
Initialize U2 GPIO low for >=10ms before enabling.
Keep high for full-scale current; do not apply ordinary PWM.
RSET gain, current drift, heat and purchased panel remain unqualified.

Circuit review in progress – not released for fabrication

Sheet: /External display, touch and backlight/
File: display.kicad_sch

Title: External display, touch and backlight

Size: A3 Date: 2026-09-24
KiCad E.D.A. 10.0.6

Rev: A – engineering draft
Id: 22/25

Always-on charging supervisor

Separate supervisor selected. U3 remains the charger. Main host powers off at SW19. Circuit capture only; routing, STM32 firmware and fault qualification pending.

A

B

C

D

E

F

A

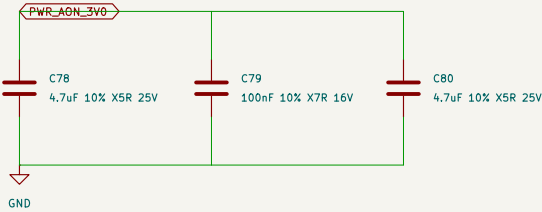
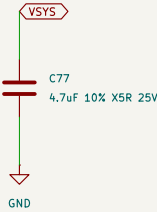
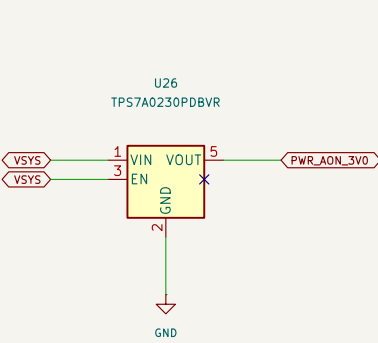
B

C

D

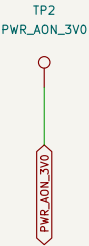
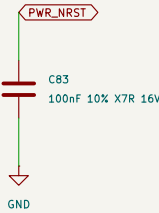
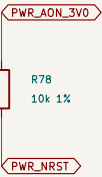
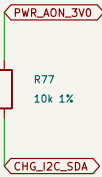
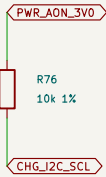
E

F



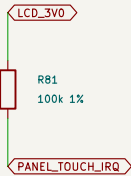
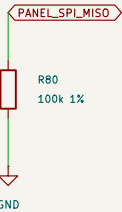
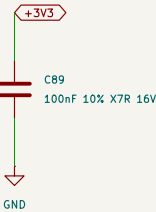
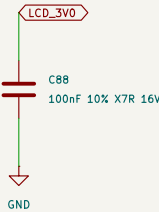
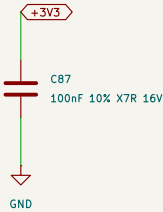
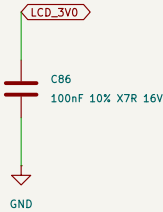
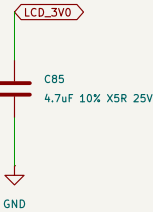
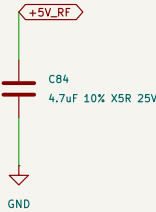
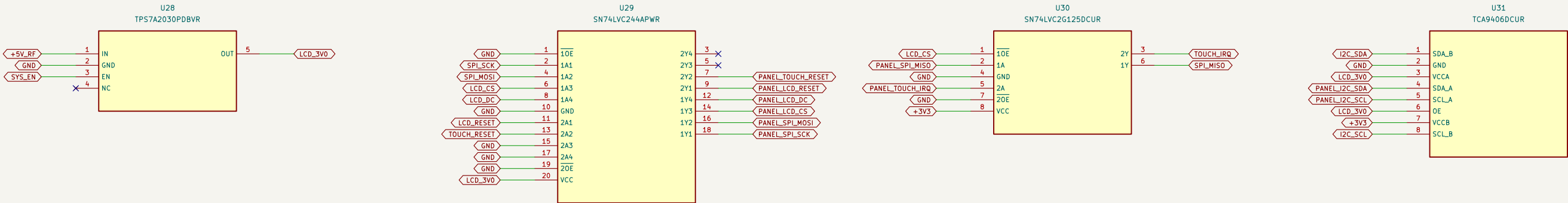
PA0: dedicated active-high charge arm. Reset analog/high-Z allows R79 at Q8 to hold charging disabled independently of U24. Never retain this output through standby or run bootloader while armed.

I2C1: PB8/PB7 AF6 -> charger 0x6A + expander 0x41.
I2C2: remap PA9/PA10 to PA11/PA12, AF6.
Host slave address 0x42 reserved; protocol not implemented.



Display supply and signal translation

Interface routed and continuity checked. Qualify loads, timing, startup and case cable. Do not fabricate.



U29: 3.3 V tolerant inputs; outputs use LCD_3V0.
U30: MISO OE follows host LCD_CS; IRQ buffer always enabled.
U31: internal pull-ups; verify bus capacitance and touch option.
U28: 5 V input provides headroom; qualify load and heat.

Circuit review in progress – not released for fabrication

Sheet: /Display supply and translation/
File: display-interface.kicad_sch

Title: Display supply and signal translation

Size: A3 Date: 2026-09-24

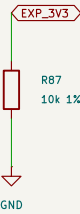
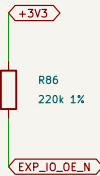
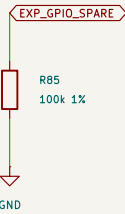
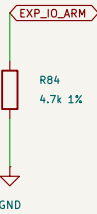
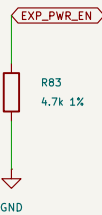
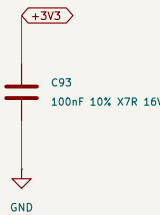
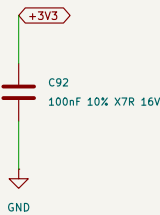
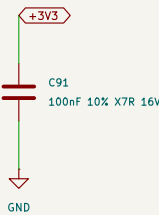
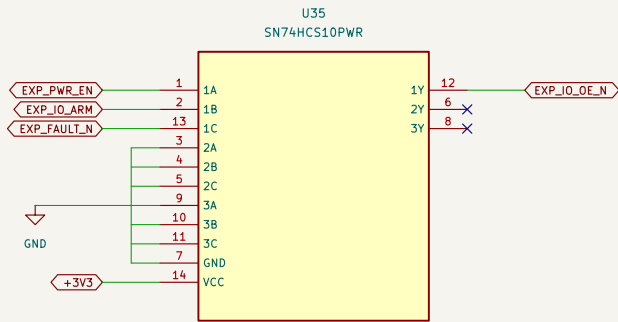
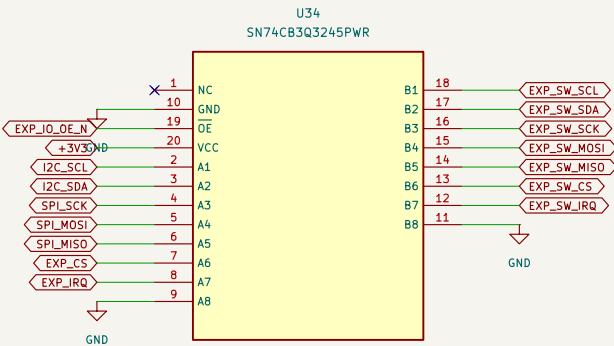
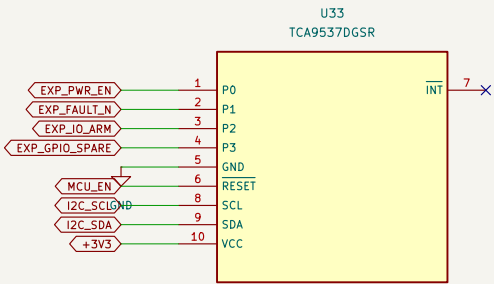
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Rev: A – engineering draft

Id: 24/25

Expansion power control and signal isolation

TCA9537 0x49: P0 power, P1 fault, P2 IO arm, P3 unused input. MCU_EN low resets all ports to inputs. External pull-downs retain OFF.



U35: /OE = NOT (EXP_PWR_EN AND EXP_IO_ARM AND EXP_FAULT_N).
Initialize output latch to 0xF0 before direction 0xF1. Power-only 0xF1: armed 0xF5.
Fault is polled; no MCU GPIO/strap changes. A stuck host I2C bus requires hardware reset.
No rail-good sensor; qualify load/startup and share buses only after settling with fault clear.
Accessory address 0x49 is reserved. Swap modules only with handset power off.

Circuit review in progress – not released for fabrication

Sheet: /Expansion control and isolation/
File: expansion-control.kicad_sch

Title: Expansion power control and signal isolation

Size: A3 Date: 2026-09-24
KiCad E.D.A. 10.0.6

Rev: A – engineering draft
Id: 25/25